nanoker

CERAMIC TECHNICAL DATA SHEET

KZM

MgO Partially Stabilized Zirconia

KEY PROPERTIES

- High thermal shock resistance
- Very low thermal conductivity
- Corrosion resistance to molten metals
- High wear resistance (sliding & abrasive)
- Good mechanical strength

APPLICATIONS

- Hot extruding dies
- Oil field components
- Valve trim
- Rollers for kilns (<1000ºC)

MAIN PROPERTIES

Property	Value	Unit
Density	5.6	g/cm^3
Flexural strength	650	MPa
Fracture Toughness	9.0	MPa·√m
Hardness	10.5	GPa
Young Modulus	210	GPa
Thermal Conductivity	3	W/m·K
CTE (20–1000ºC)	7.5	$x10^{-6}$ K^{-1}
Electrical Resistivity	$>10^{11}$	Ω·cm

** All properties measured at 20ªC unless othervise stated.*
*** Measured in a wall thickness of 2 mm*

OPERATING CONDITIONS

Max working temp (air)	1000	ºC
Recomended temp	<900	ºC
Continuous temp	850	ºC

ELECTRICAL BEHAVIOR

Insulator	
Electrical resistivity	$>10^{11}$ Ω·cm

MATERIAL INFORMATION

Material type	MgO stabilized zirconia
Structure	Partially stabilized ceramic
Key feature	High toughness
Machinability	Medium
Color	Black

Local overheating may occur at electrical contact points due to current concentration and geometry effects.

Nanoker Research S.L.
Polígono de Olloniego, Parcela 21A, Nave 4-8, 33660, Oviedo (Asturias), Spain
(+34) 985 20 76 13 | info@nanoker.com

Version 5.0 (Mar. 2026)